

ACTIVE PROBE With High Input Impedance And Selectable Gain

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A high input impedance AC and DC active probe is required to increase the input resistance and sensitivity of such measuring instruments as oscilloscopes and multimeters. Presented here is one such solution that consists of a single operational amplifier (op-amp) and selectable gain. The active probe protects inputs of the test equipment during testing and experimentation. The probe can easily connect to and disconnect from other equipment.

Circuit and working

The circuit diagram of the simple AC and DC active probe with high input impedance and selectable gain is shown in Fig. 1. It is built around an op-amp TL071 (IC1) and a few other components.

The circuit is designed for a single op-amp with a stable gain of +1 and low input current (JFET inputs). It is capable of driving at least 600-ohm load. Op-amps like OPA132, OPA134, TLE2071 and TLE2081 may be used here. You may also use MCP601, but the power supply should be $\pm 3V$. In fact, you can use a large variety of op-amps with JFET inputs, including the classical TL061, TL071, TL081, LF356, LF357 and so on, with slight modifications including resistors, capacitors and corresponding power supplies.

Switch S1 is closed for DC input signals and it is open when you wish to work only with AC input signals.

To make it simple and compact, the probe does not use a multi-trimmer potentiometer or any additional components for adjusting offset of the op-amp. Including offset is not a big issue; it can be done at the output of the probe. Usually, offset does not change much if you use a regulated low-noise power supply and ambient temperature does not change a lot.

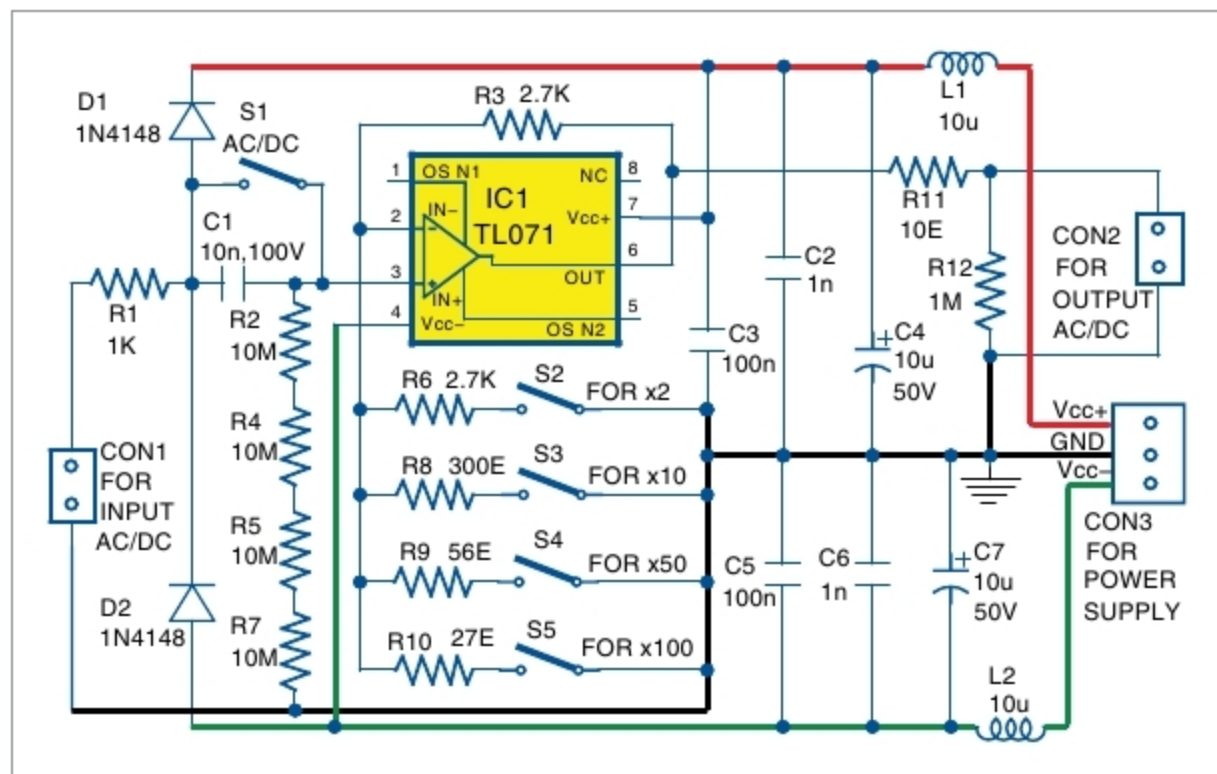


Fig. 1: Circuit diagram of active probe

SELECTING GAIN OF THE ACTIVE PROBE

Switch position				Gain
S2	S3	S4	S5	
Open	Open	Open	Open	+1
Closed	Open	Open	Open	+2
Open	Closed	Open	Open	+10
Open	Open	Closed	Open	+50
Open	Open	Open	Closed	+100

Input resistance. Input resistance of the probe is set to 40-mega-ohm using resistors R2, R4, R5 and R7. Such high input impedance makes the probe appropriate for examining signal sources with high internal resistance and with low level of output signals.

Selecting the gain. The probe does not need any adjustment to work properly. Before using the probe, set the position of S1 (AC or DC input) and the gain (close one of the switches from S2 to S5). Gain of the active probe is set with the four switches S2 through S5, as described in the table. Bandwidth

PARTS LIST

Semiconductors:

- IC1 - TL071 op-amp
- D1-D2 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2, R4, R5, R7, R12 - 10-mega-ohm
- R3, R6 - 2.7-kilo-ohm
- R8 - 300-ohm
- R9 - 56-ohm
- R10 - 27-ohm
- R11 - 10-ohm

Capacitors:

- C1 - 10nF, 100V ceramic disk
- C2, C6 - 1nF ceramic disk
- C3, C5 - 100nF ceramic disk
- C4, C7 - 10uF, 50V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin connector
- CON3 - 3-pin connector
- S1-S5 - On/off switch
- 8-pin base
- Dual power supply/two 9V batteries 6F22

depends on the actual op-amp used.

It is preferable to select resistors R3, R6, R8 to R10 from E96 series. You can change the gain to any appropriate value as per requirement.

Power supplies. This circuit